



ya diggg: Project Scope & Technical Constraints

This document defines the boundaries of the ya diggg project to ensure technical focus on the "Music Intelligence" mission and maintain the "Disappearing Device" philosophy.

1. Core Feature Scope (IN-SCOPE)

A. Music Intelligence

- **Offline OCR Engine:** Fast recognition of artist, album, and catalog metadata from covers/labels.
- **Local Metadata Lookup:** On-device database containing credits, personnel, studios, and sample lineages.
- **"Facts-Only" Narrative:** Strict presentation of data (Personnel, Labels, Dates) without subjective rankings or recommendations. [Youtube](#)

B. Hardware Interaction

- **One-Button Scan:** Single-press triggers capture, OCR, and reveal.
- **Tactile Scrolling:** Mechanical dial/toggle for browsing long personnel lists.
- **Sunlight Readability:** 1-bit monochrome display optimized for record fair environments. [Stackexchange](#)
- **12+ Hour Active Battery:** Enough for a full day of "crate digging."

C. Data Management

- **USB-C Library Sync:** One-way sync from PC/Mac to update the local database.
- **Offline Storage:** Minimum 32GB eMMC to store the music database and user "scanned" history.

2. Excluded Features (OUT-OF-SCOPE)

Feature	Exclusion Rationale
WiFi / Bluetooth	Increases complexity, reduces battery life, and violates the "Offline Soul" philosophy.
Social Media Integration	The device is for private discovery, not public sharing.
Music Streaming / Playback	The record is the hero; ya diggg does not compete with the listening experience.
Rankings / Scores	Violates the principle of being a neutral "companion." No ratings from Discogs, Pitchfork, etc.
Marketplace / Buying	No direct "Buy on Discogs" links. The device reveals facts, not commercial opportunities.

3. Technical Constraints

- **Latency Target:** < 1.5 seconds from Button Press to Metadata Reveal.
- **Z-Height Constraint:** Maximum device thickness of 12mm to maintain "pocketable" status.
- **Database Capacity:** Support for at least 1,000,000 master record indices locally (compressed lookup tables).
- **OCR Accuracy:** > 95% on high-contrast center labels; > 85% on small-print archival credits.

4. Scope Governance

- **Feature Creep:** Any request for "Smart" features (voice assistants, notifications, apps) is automatically rejected to maintain device purity.
- **Updates:** Database updates are handled exclusively via wired USB-C connection to ensure privacy and simplicity.